

Project Design Phase

Problem – Solution Fit Template

Date	19-02-2026
Team ID	LTVIP2026TMIDS88080
Project Name	Heart Disease Analysis: Identifying Risk Factors Using Tableau.
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem–Solution Fit means identifying a real healthcare data analysis problem and ensuring that the developed analytical solution effectively addresses it. In this project, it ensures that the Heart Disease Analysis dashboard solves the difficulty faced by healthcare stakeholders in understanding complex medical data.

It helps healthcare analysts, researchers, and public health officials recognize patterns, risk factors, and trends in heart disease data through interactive visualization.

Purpose

- ☐ Solve complex healthcare data analysis problems in a way that fits the needs of healthcare stakeholders.
- ☐ Improve decision-making speed and effectiveness by providing interactive dashboards aligned with user behaviour (filtering, comparison, drill-down analysis).
- ☐ Enhance communication of medical insights through clear visual storytelling and data-Driven messaging.
- ☐ Increase stakeholder engagement by addressing critical health concerns such as high-risk groups, lifestyle factors, and demographic trends.
- ☐ Understand the current heart disease patterns and risk factors in order to support better healthcare planning and preventive strategies.

Template:

STAKEHOLDER ISSUES	1. TARGET USERS / STAKEHOLDERS (TS) TS - Healthcare Analysis - Medical Researchers - Public Health Officials - Hospitals & Healthcare Institutions - Policy Makers	6. PROJECT CONSTRAINTS (PC) AS - Limited real-time medical data - Data privacy considerations - Dataset size limitations - Dependence on provided dataset	5. EXISTING METHODS OF ANALYSIS (AS) AS - Manual data reports (Excel) - Static charts and summary statistics - Traditional research papers - Basic hospital record analysis
	2. HEALTHCARE PROBLEMS TO ANALYZE (HP) HP - Identify heart disease prevalence patterns - Analyze risk factors (age, gender, diabetes, smoking, stroke) - Compare demographic distributions - Detect high-risk population groups	9. ROOT CAUSES OF HEART DISEASE (RC) RC - Lifestyle factors (smoking, alcohol, inactivity) - Chronic conditions (diabetes, hypertension) - Age-related risks - Lack of preventive care	7. USER INTERACTION WITH DASHBOARD (UI) UI - Filter by age, gender, race - Compare diabetic vs non-diabetic patients - Analyze impact of smoking & alcohol - Explore interactive dashboards and story points
TRIGGERS	3. ANALYSIS TRIGGERS (AT) AT - Rising heart disease cases - Need for preventive healthcare insights - Requirement for data-driven-policy decisions - Research and academic analysis needs	7. FACTORS SPOOT CAUSE (AT) - Rising heart disease cases - Need for preventive healthcare insights - Requirement for data-driven-policy decisions - Research and academic analysis needs	7. USER INTERACTION WITH DASHBOARD (UI) UI - Filter by age, gender, race - Compare diabetic vs non-diabetic patients - Analyze impact of smoking & alcohol - Explore interactive dashboards and story points
ACCESS CHANNELS	4. STAKEHOLDER EXPECTATIONS (BEFORE/AFTER ANALYSIS) BE Before: Limited visibility into risk patterns After: Clear insights, better decision-making, improved healthcare planning	10. HEART DISEASE ANALYSIS SOLUTION (HS) HS - Import provided dataset (CSV) into Tableau - Perform data cleaning & transformation - Create interactive dashboards and stories - Deliver a scalable and reusable analytical model	8. ACCESS CHANNELS (AC) AC ONLINE: Tableau Public dashboards, shared Links OFFLINE: Project reports, presentations, academic submissions
	4. STAKEHOLDER EXPECTATIONS (BEFORE/AFTER ANALYSIS) ST Before: Limited visibility into risk patterns After: Clear insights, better decision-making, improved healthcare planning		

AS THE TEMPLATE DOWNLOADED FROM THE GIVEN RESOURCE LINK ALL ARE IN PNG FORM
SO JUST MADE IT INTO TABLE FORMAT:

Section	Solution
Identified Problem	Healthcare analysts and researchers find it difficult to interpret heart disease trends due to raw, unstructured data without clear visual representation.
Proposed Solution	Develop interactive charts, dashboards, and storyboards using Tableau to clearly visualize heart disease prevalence, risk factors (age, gender, diabetes, smoking, stroke), and demographic patterns.
Why This Solution?	It transforms raw heart disease data into easy-to-understand visual insights. Users can apply filters, compare categories, and identify high-risk groups without manual data analysis.
Fit for Target User?	Yes, healthcare analysts, researchers, and public health officials receive a clear, interactive, and professional visualization that supports faster and more informed decision-making.

